Validity Analysis: FOLIO

Target context: US AI Research Lab — FOLIO FOL Reasoning Evaluation

Overall risk: **LOW**

Deployment Context

We are a US AI research lab evaluating LLM reasoning capabilities. We want to test whether the model can correctly classify the logical relationship (entailment, contradiction, or neutral) between a set of premises and a conclusion expressed in formal logical statements. Input is structured premise-conclusion pairs. Output is a categorical label. Our evaluation metric is classification accuracy.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	5	Strong alignment	high
Input Content	4	Minor gaps	high
Input Form ■	3	Moderate gaps	high
Output Ontology ✓	5	Strong alignment	high
Output Content ✓	4	Minor gaps	high
Output Form	4	Minor gaps	high
Average	**4.2**		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

FOLIO is a strongly aligned benchmark for the US AI research lab's classical FOL reasoning evaluation, with five of six validity dimensions scoring 4–5. Input ontology and output ontology are exact matches: FOLIO's classical FOL operator coverage [Q138] and explicit temporal/modal exclusions [Q139, Q140] correspond precisely to the lab's in-scope and out-of-scope reasoning types, and the True/False/Unknown taxonomy maps cleanly onto entailment/contradiction/neutral per elicitation Q1. Output content is anchored in objective FOL-inference-derived labels [Q2, Q153], sidestepping most annotator-subjectivity concerns. The primary live concern, consistent with the lab's HIGH priority on input form, is the unresolved choice between pure-symbolic and hybrid NL+FOL presentation — FOLIO was designed and validated for NL-first presentation, and the paper itself documents that input format materially affects model scores [Q162, Q163].

Dataset analysis surfaces concrete annotation quality concerns: a direct NL/FOL semantic contradiction (Example 57, negation dropped) [DATASET-D17], multiple syntactic errors in FOL formulas, entity-name inconsistencies, and FOL over-specification patterns — corroborated by an independent 2025 study using the Vampire prover that established a cleaned validation set [WEB-5]. Output-form concerns are moderate: the HuggingFace dataset lacks the documented test split, the original GPT-4 ceiling has been substantially closed by neuro-symbolic pipelines (88.32% SotA) [WEB-4], and systematic label-class asymmetry [Q155-Q157] means aggregate accuracy can mislead without per-class reporting.

ID	Evidence
Q1	"FOLIO consists of 1,430 examples (unique conclusions), each paired with one of 487 sets of premises used to deductively reason for the validity of each conclusion." (p.1)
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q138	"We include the following operators: negation ¬, conjunction ∧, disjunction ∨, implication →, universal quantifier ∀, existential quantifier ∃, equal =." (p.13)
Q139	"Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics." (p.13)
Q140	"Consequently, they are beyond the scope of the definition of first-order logic used in our dataset." (p.13)
Q153	"Proving a story requires three steps. First, the FOL statements of the premises and conclusions of a story annotated by humans are converted to Python code. Then, the code snippets are used as input to the theorem prover. Finally, the theorem prover outputs whether the conclusions are True / False / Unknown, based on the premises." (p.14)
Q155	"Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown." (p.15)
Q156	"The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting." (p.15)
Q157	"They also tend to make more predictions of True than the other labels." (p.15)
Q162	"As shown in Table 11, the performance slightly increases in the NL+FOL setting for GPT-4 while GPT-3.5 performs worse in both the NL+FOL and the FOL-only settings." (p.15)
Q163	"FOL always serves as additional useful information for GPT-4, but not for GPT-3.5 regardless of whether FOL is concatenated with NL." (p.15)
D17	Matt does not invest in the public stock market regularly." / "InvestInRegularly(matt, publicStockMarket) — NL premise says Matt does NOT invest; FOL asserts he DOES — direct NL/FOL semantic contradiction
WEB-4	LLM-ARC neuro-symbolic system achieves 88.32% on FOLIO validation (2024 SotA) (https://arxiv.org/pdf/2406.17663)
WEB-5	Auto-formalization agent 86.70% on cleaned FOLIO vs. 73.89% CoT baseline (https://arxiv.org/pdf/2603.02788)

Practical Guidance

What This Benchmark Measures

For this lab's deployment, FOLIO provides a high-fidelity evaluation of classical first-order logic reasoning capability — specifically deductive entailment over premise-conclusion pairs with quantifiers, negation, conditionals, conjunction/disjunction, and multi-step chains (mode-4 depth, 28.7% at depth ≥5). The input_ontology (score 5) and output_ontology (score 5) dimensions show exact construct alignment, and output_content (score 4) confirms labels are anchored in objective FOL inference rather than subjective judgment. The benchmark cleanly supports the lab's three-way accuracy metric.

Construct Depth

Construct depth is strong for the primary classification task and weaker for the symbolic-input regime. The NL-first design [Q64-Q68], NL-FOL alignment conventions [Q130-Q133], and per-sentence parallel annotation [DATASET-D1, D2] provide rich signal for hybrid evaluation. However, three depth limitations apply: (1) per-depth statistical power is limited at depth 6–7 (likely single-digit test examples) [WEB-6], (2) WikiLogic real-world entity references create a documented shortcut-reasoning confound [Q104, DATASET-D8, D10] — subcorpus-stratified analysis is needed, and (3) approximately 1–5% of sampled examples show NL/FOL annotation defects that may corrupt training signal for the secondary translation task [DATASET-D14-D17, D20-D21].

What Else You Need

For the lab's specific use case: (1) If pursuing pure-symbolic evaluation (the input_form score-3 concern), supplement FOLIO with ProverQA [WEB-2] or LogicGraph [WEB-3], which are designed for symbolic-input regimes with depth stratification. (2) Verify whether to use the original or 2025-cleaned FOLIO annotations [WEB-5] — this directly affects output_content reliability. (3) Source the 226-example test split from FOLIO GitHub rather than HuggingFace (test split missing from HF). (4) Run subcorpus-stratified analysis (WikiLogic vs. HybLogic) to disentangle contamination effects from genuine deductive capability. (5) For frontier-model discriminability (output_form), supplement with LogiConBench [WEB-9] if FOLIO appears saturated under neuro-symbolic pipelines.

ID	Evidence
Q64	“Each natural language (NL) story S in FOLIO consists of n premises: P = {p1, p2, ..., pn} and m conclusions: H = {h1, h2, ..., hm}.” (p.6)
Q65	“All NL stories are annotated with parallel FOL stories SF, which are sets of FOL formulas consisting of n premises PF = {pf1, pf2, ..., pfn} and m conclusions HF = {hf1, hf2, ..., hfm}.” (p.6)
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q67	“We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset.” (p.6)
Q68	“In particular, each of the NL sentence pi or hi and the parallel FOL formula pfi or hfi should be logically and semantically equivalent.” (p.6)
Q104	“Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data.” (p.8)
Q130	“We always use "either-or" to express exclusive disjunction.” (p.13)
Q131	“We use either "A or B" or "A or B, or both" to express inclusive disjunction.” (p.13)
Q132	“It is more natural to say "Some A is B" rather than "there exists an A such that A is B."” (p.13)
Q133	“"All A are B" can be more natural than "If A then B".” (p.13)
D1	No plants are fungi. Mushrooms are fungi." / "∀x (Plant(x) → ¬Fungi(x)) ∀x (Mushroom(x) → Fungi(x)) — Simple 2-premise syllogism with exact NL/FOL parallel; confirms NL-first design with FOL annotation layer
D2	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises." / "∀x (InThisTechCompany(x) ∧ Consistent(x) ∧ StickTo(x, theirRegularRoutine) → ¬Like(x, surprise)) — Multi-premise story with 7 premises including mixed disjunction/implication; illustrates chain-depth complexity and NL-FOL alignment
D8	Oxford Circus is the entrance to Oxford Circus tube station, a part of the Central line in 1900." / "EntraceTo(oxfordCircus, tubeStation) ∧ PartOf(tubeStation, centralline) ∧ In(tubeStation, 1900) — WikiLogic story with real-world entity (Oxford Circus, John Nash); typo "EntraceTo" in FOL
D10	Michael O'Donnell is a British physician, journalist, author, and broadcaster." / "British(michael) ∧ Physician(michael) ∧ Journalist(michael) ∧ Author(michael) ∧ Broadcaster(michael) — WikiLogic: real Wikipedia person; FOL uses "michael" as constant but full name is John Nash / Michael O'Donnell
D14	Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic error in training data
D15	∀x (MajorArgumentForm(x) → (InductiveReasoning(x) ⊕ DeductiveReasoning(x)) — Missing closing parenthesis in premises-FOL
D16	Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises." / "¬(ProfessionalBasketballPlayer(jim) ⊕ NeverExercises(jim)) — Premises describe "John" but FOL uses "jim" — name inconsistency between NL and FOL
D17	Matt does not invest in the public stock market regularly." / "InvestInRegularly(matt, publicStockMarket) — NL premise says Matt does NOT invest; FOL asserts he DOES — direct NL/FOL semantic contradiction
D20	(Knows(dan, dune) ∧ ScienceFiction(dune)) ∨ ProvedToBe(dune, false)) — Unbalanced parenthesis in premises-FOL
D21	Contain(tikTok, chatFeature) ⊕ ComputerProgram(tikTok)) — Unbalanced closing parenthesis in conclusion-FOL
WEB-2	ProverQA (ICLR 2025) — pure-symbolic-compatible alternative if lab pursues that regime (https://proceedings.iclr.cc/paper_files/paper/2025/file/862819c227b16f9af64dd6ad6cfdf275-Paper-Conference.pdf)
WEB-3	LogicGraph — depth-stratified symbolic FOL benchmark (https://arxiv.org/html/2602.21044v1)

WEB-5	Auto-formalization agent 86.70% on cleaned FOLIO vs. 73.89% CoT baseline (https://arxiv.org/pdf/2603.02788)
WEB-6	FOLIO depth range 0–7, mode=4, 28.7% at depth ≥5 confirmed in EMNLP 2024 version (https://aclanthology.org/2024.emnlp-main.1229.pdf)
WEB-9	LogiConBench designed to remain discriminative when existing benchmarks saturate (https://openreview.net/pdf?id=ULEHJkolxB)

Input Ontology (IO)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US AI Research Lab — FOLIO FOL Reasoning Evaluation*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's taxonomy aligns precisely with the lab's stated scope of classical deductive FOL. The benchmark covers all standard FOL operators (negation, conjunction, disjunction, implication, universal and existential quantifiers, equality) [Q138], uses n-place predicates for expressivity [Q141], explicitly excludes temporal and modal logic as out-of-scope [Q139, Q140] — matching the lab's out-of-scope list exactly — and emphasizes multi-step reasoning depth (mode=4, 28.7% at depth ≥5) [Q60]. The two-task structure (NL reasoning + NL-FOL translation) [Q63, Q67] covers both the lab's primary classification task and a documented secondary interest. Dataset analysis confirms all core operators are present in actual data, including exclusive disjunction (⊕) and quantifiers [DATASET-D3, D9, D13]. One minor caveat: per-depth example counts at depth 6–7 are likely too small for reliable per-depth statistical inference [WEB-6], but this affects statistical power, not taxonomic coverage.

ID	Evidence
Q60	“As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1.” (p.5)
Q63	“We define two new tasks based on FOLIO, natural language reasoning with first-order logic and NL-FOL translation.” (p.6)
Q67	“We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset.” (p.6)
Q138	“We include the following operators: negation ¬, conjunction ∧, disjunction ∨, implication →, universal quantifier ∀, existential quantifier ∃, equal =.” (p.13)
Q139	“Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics.” (p.13)
Q140	“Consequently, they are beyond the scope of the definition of first-order logic used in our dataset.” (p.13)
Q141	“We use n-place predicates when applicable for the expressivity of the FOL formulas.” (p.13)
D3	A project is written either in C++ or Python." / "∀x (Project(x) → (WrittenIn(x, cplusplus) ⊕ WrittenIn(x, python))) — Exclusive disjunction (⊕) expressed in NL as "either-or"; confirms documented NL convention alignment
D9	All professional athletes spend most of their time on sports... Amy spends the most time on sports, or Amy is an Olympic gold medal winner." / "∀x (ProfessionalAthlete(x) → SpendOn(x, mostOfTheirTime, sports)) ... SpendOn(amy, mostOfTheirTime, sports) ∨ OlympicGoldMedalWinner(amy) — Multi-step deductive chain with negation and implication; 6 premises, illustrates chain-depth coverage
D13	If PSO J318.5–22 is not both a rogue planet and a planet gravitationally bound by the Sun, then it is a rogue planet." / "¬(Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22) ∧ BoundBy(pSOJ318.5-22, sun, gravitationally)) → (Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22)) — Complex compound premise with negation and conditional; WikiLogic astronomy topic
WEB-6	FOLIO depth range 0–7, mode=4, 28.7% at depth ≥5 confirmed in EMNLP 2024 version (https://aclanthology.org/2024.emnlp-main.1229.pdf)

Strengths

- Full classical FOL operator coverage verified in both documentation and sampled data
- Explicit exclusion of out-of-scope sub-types (temporal, modal) matches the lab's scope precisely
- Substantially deeper reasoning-chain coverage (28.7% at depth ≥5) than prior FOL benchmarks (RuleTaker, LogicNLI)

• Multi-conclusion-per-premise design tests world-model consistency, not just isolated inference

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Lab requires classical deductive FOL: quantifiers, negation, conditionals, conjunction/disjunction, multi-step deduction. All confirmed present in FOLIO documentation [Q138] and sampled data [DATASET-D3, D9, D13].

IO-2: No regionally-relevant categories omitted. The lab's in-scope list (classical deductive FOL) is precisely what FOLIO covers; out-of-scope items (temporal, modal, arithmetic, probabilistic) are explicitly excluded by FOLIO [Q139, Q140] and confirmed out-of-scope by the lab.

IO-3: FOLIO does not include irrelevant categories beyond its declared scope; the NL-FOL translation secondary task is documented as a separate task with its own metrics [Q63, Q67, Q73-Q76], so it does not contaminate the primary classification metric.

IO-4: No content-validity gaps in the taxonomy. Statistical-power concern at depth 6–7 is documented but is a sample-size issue, not a taxonomic gap [WEB-6].

ID	Evidence
Q63	“We define two new tasks based on FOLIO, natural language reasoning with first-order logic and NL-FOL translation.” (p.6)
Q67	“We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset.” (p.6)
Q73	“Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV).” (p.6)
Q74	“The Syntactic Validity score measures whether the FOL formulas are syntactically valid. The score will be 1 if all FOL formulas of an example can pass the syntactic check and 0 otherwise 2).” (p.6)
Q75	“Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion.” (p.6)
Q76	“We define the accuracy of the output labels as the execution accuracy.” (p.6)
Q138	“We include the following operators: negation ¬, conjunction ∧, disjunction ∨, implication →, universal quantifier ∀, existential quantifier ∃, equal =.” (p.13)
Q139	“Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics.” (p.13)
Q140	“Consequently, they are beyond the scope of the definition of first-order logic used in our dataset.” (p.13)
D3	A project is written either in C++ or Python." / "∀x (Project(x) → (WrittenIn(x, cplusplus) ⊕ WrittenIn(x, python))) — Exclusive disjunction (⊕) expressed in NL as "either-or"; confirms documented NL convention alignment
D9	All professional athletes spend most of their time on sports... Amy spends the most time on sports, or Amy is an Olympic gold medal winner." / "∀x (ProfessionalAthlete(x) → SpendOn(x, mostOfTheirTime, sports)) ... SpendOn(amy, mostOfTheirTime, sports) ∨ OlympicGoldMedalWinner(amy) — Multi-step deductive chain with negation and implication; 6 premises, illustrates chain-depth coverage
D13	If PSO J318.5–22 is not both a rogue planet and a planet gravitationally bound by the Sun, then it is a rogue planet." / "¬(Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22) ∧ BoundBy(pSOJ318.5-22, sun, gravitationally)) → (Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22)) — Complex compound premise with negation and conditional; WikiLogic astronomy topic
WEB-6	FOLIO depth range 0–7, mode=4, 28.7% at depth ≥5 confirmed in EMNLP 2024 version (https://aclanthology.org/2024.emnlp-main.1229.pdf)

Information Gaps

- Per-depth example counts at depth 6–7 in the test set not published; depth 6–7 subsets likely single-digit counts

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Remediation

Gap 1: Per-depth example counts at depth 6–7 are likely single-digit in the test set, limiting statistical reliability of per-depth performance estimates

Recommendation: When reporting per-depth performance, aggregate to depth bands (0–3, 4–5, 6+) rather than per-depth points, and provide confidence intervals. For finer-grained depth analysis, supplement with ProverQA which provides 500 examples per difficulty band.

Input Content (IC)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US AI Research Lab — FOLIO FOL Reasoning Evaluation*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content is well-aligned for the lab's expert evaluator population in a US research lab context. FOLIO content consists of logically structured fictional stories with no consumer-facing cultural-loading concerns, expert-annotated by CS students/researchers [Q16, Q31] with explicit bias-screening (race, ethnicity, gender, sexuality, nationality, class, religion excluded) [Q47, Q127] and factual accuracy review [Q126]. The 4,351-word vocabulary spans diverse real-world topics [Q62], confirmed empirically across astronomy, geography, biology, economics, and media in the sample [DATASET-D13, D26, D28]. The primary concerns are (1) WikiLogic real-world entity references may overlap with LLM pretraining data, creating a shortcut-reasoning confound [Q104, DATASET-D8, D10] — explicitly flagged in the paper; and (2) NL-layer typos and grammar issues observed in training data despite documented quality control [DATASET-D7, D27]. Neither concern is fatal for an expert evaluator population that can perform subcorpus-stratified analysis (HybLogic vs. WikiLogic) [Q102].

ID	Evidence
Q16	“FOLIO is a high-quality and manually curated dataset, written by CS undergraduate and graduate students and researchers in academia and industry.” (p.2)
Q31	“All stories and FOL annotations in FOLIO are written and reviewed by expert annotators, including CS undergraduate and graduate students, and senior researchers, who met the aforementioned criteria.” (p.4)
Q47	“Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion.” (p.4)
Q62	“Table 3 shows that our dataset has a vocabulary of 4,351 words, and the examples based on Wikipedia account for 74% of the total vocabulary even though the WikiLogic stories take up only 63% of the total number of stories.” (p.5)
Q102	“As shown in Table 6, in logical reasoning, GPT-3.5 and GPT-4 achieve substantially lower results on HybLogic than on WikiLogic and the result is slightly higher than chance.” (p.8)
Q104	“Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data.” (p.8)
Q126	“We rewrote those that are not reflective of well-established scientific, historical, or legal facts.” (p.13)
Q127	“We took out stories that had strongly opinionated language and contained gender, racial, and classist biases.” (p.13)
D7	No satin-finish lipsticks in the set do not have 'rosewood' in its offical description. — Double-negative NL sentence with typo ("offical"); NL quality issue in training data
D8	Oxford Circus is the entrance to Oxford Circus tube station, a part of the Central line in 1900." / "EntraceTo(oxfordCircus, tubeStation) ∧ PartOf(tubeStation, centralline) ∧ In(tubeStation, 1900) — WikiLogic story with real-world entity (Oxford Circus, John Nash); typo "EntraceTo" in FOL
D10	Michael O'Donnell is a British physician, journalist, author, and broadcaster." / "British(michael) ∧ Physician(michael) ∧ Journalist(michael) ∧ Author(michael) ∧ Broadcaster(michael) — WikiLogic: real Wikipedia person; FOL uses "michael" as constant but full name is John Nash / Michael O'Donnell
D13	If PSO J318.5–22 is not both a rogue planet and a planet gravitationally bound by the Sun, then it is a rogue planet." / "¬(Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22) ∧ BoundBy(pSOJ318.5-22, sun, gravitationally)) → (Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22)) — Complex compound premise with negation and conditional; WikiLogic astronomy topic
D26	Ho is at the business conference and prefers state ownership of the means of production. — Non-Western name ("Ho") in story; ideologically loaded topic (communist/planned economy) used as logical content — within scope of deductive task

D27	Lipstcks in the Rouge Dior set, Lunar New Year Limited Edition either does not have 'rosewood' in its offical description or it has 'rosewood' in its official description. — Typo "Lipstcks" in NL premises; tautological premise (p ∨ ¬p)
D28	Xiufeng, Xiangshan, Diecai, Qixing are districts in the city of Guilin." / "DistrictIn(xiufeng, guilin) ∧ DistrictIn(xiangshan, guilin)... — WikiLogic: Chinese geographic entities (Guilin districts); conclusion about Kowloon/HK is unrelated to premises — exemplifies "Unknown" label for unrelated conclusions

Strengths

- Identity-based bias screening was performed and 39.2% of stories rewritten [Q48, Q127]
- Broad topical coverage in sampled data confirms documented vocabulary breadth [DATASET-D13, D26, D28]
- Expert annotation with FOL training reduces semantic-content noise [Q31]
- Cultural loading is irrelevant to this expert US AI lab deployment per elicitation

ID	Evidence
Q31	“All stories and FOL annotations in FOLIO are written and reviewed by expert annotators, including CS undergraduate and graduate students, and senior researchers, who met the aforementioned criteria.” (p.4)
Q48	“Toward these objectives, we manually screened all stories and found that 39.2% of the stories suffer from at least one of these issues. We implemented a detailed protocol to rewrite these stories.” (p.4)
Q127	“We took out stories that had strongly opinionated language and contained gender, racial, and classist biases.” (p.13)
D13	If PSO J318.5–22 is not both a rogue planet and a planet gravitationally bound by the Sun, then it is a rogue planet." / "¬(Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22) ∧ BoundBy(pSOJ318.5-22, sun, gravitationally)) → (Planet(pSOJ318.5-22) ∧ Rogue(pSOJ318.5-22)) — Complex compound premise with negation and conditional; WikiLogic astronomy topic
D26	Ho is at the business conference and prefers state ownership of the means of production. — Non-Western name ("Ho") in story; ideologically loaded topic (communist/planned economy) used as logical content — within scope of deductive task
D28	Xiufeng, Xiangshan, Diecai, Qixing are districts in the city of Guilin." / "DistrictIn(xiufeng, guilin) ∧ DistrictIn(xiangshan, guilin)... — WikiLogic: Chinese geographic entities (Guilin districts); conclusion about Kowloon/HK is unrelated to premises — exemplifies "Unknown" label for unrelated conclusions

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Lab deployment is a US AI research lab with English-only content and expert evaluators; no region-specific cultural/geographic/dialectal knowledge requirements [elicitation cultural_norms_notes]. FOLIO's Wikipedia-seeded content spans diverse topics but does not require non-Western knowledge.

IC-2: FOLIO explicitly excludes identity-based biases [Q47, Q127]; content is logical-structure-focused rather than culturally normative. Alignment with US research lab context is strong.

IC-3: WikiLogic references real-world entities (Oxford Circus, Michael O'Donnell, John Nash) [DATASET-D8, D10] — these are Western-centric but the deployment is US-based, so this is a population match. However, real-world entity references create a documented contamination/shortcut-reasoning confound [Q104].

IC-4: Not applicable — expert AI researcher population in target context; regional annotator recruitment irrelevant per elicitation.

IC-5: Content issues identified: (a) WikiLogic contamination risk [Q104], (b) typos and grammar issues in training data observed in sample [DATASET-D7, D27], (c) tautological premise in Example 13 [DATASET-D27]. None are fatal but introduce construct-irrelevant variance.

ID	Evidence
Q47	"Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion." (p.4)
Q104	"Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data." (p.8)
Q126	"We rewrote those that are not reflective of well-established scientific, historical, or legal facts." (p.13)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)
D7	No satin-finish lipsticks in the set do not have 'rosewood' in its offical description. — Double-negative NL sentence with typo ("offical"); NL quality issue in training data
D8	Oxford Circus is the entrance to Oxford Circus tube station, a part of the Central line in 1900." / "EntraceTo(oxfordCircus, tubeStation) ∧ PartOf(tubeStation, centralline) ∧ In(tubeStation, 1900) — WikiLogic story with real-world entity (Oxford Circus, John Nash); typo "EntraceTo" in FOL
D10	Michael O'Donnell is a British physician, journalist, author, and broadcaster." / "British(michael) ∧ Physician(michael) ∧ Journalist(michael) ∧ Author(michael) ∧ Broadcaster(michael) — WikiLogic: real Wikipedia person; FOL uses "michael" as constant but full name is John Nash / Michael O'Donnell
D27	Lipstcks in the Rouge Dior set, Lunar New Year Limited Edition either does not have 'rosewood' in its offical description or it has 'rosewood' in its official description. — Typo "Lipstcks" in NL premises; tautological premise (p ∨ ¬p)
WEB-5	Auto-formalization agent 86.70% on cleaned FOLIO vs. 73.89% CoT baseline (https://arxiv.org/pdf/2603.02788)

Information Gaps

- Rate of NL quality issues across the full corpus cannot be estimated from 57-example sample
- Whether lab uses cleaned (2025) or original FOLIO annotation set not specified

Requires Expert Verification

- Whether contamination from Wikipedia-seeded stories materially affects model rankings for the lab's specific model set

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Remediation

Gap 1: WikiLogic real-world entity references create contamination/shortcut-reasoning confound; HybLogic vs. WikiLogic subcorpus tag is not in the HuggingFace schema

Recommendation: Cross-reference the FOLIO GitHub repository to tag each example with its subcorpus origin. Report HybLogic and WikiLogic accuracies separately to disentangle genuine deductive capability from world-knowledge shortcuts.

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US AI Research Lab — FOLIO FOL Reasoning Evaluation*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is the lab's highest-priority dimension and the analysis surfaces real concerns despite strong native-format alignment. FOLIO is fundamentally NL-first with parallel FOL annotations [Q64, Q65, Q37]; NL and FOL are designed to be logically and semantically equivalent [Q68], grammar-reviewed [Q49, Q50], and follow standard AI/education FOL syntax [Q52]. The parallel structure is empirically confirmed at sentence-level granularity [DATASET-D1, D2], and documented NL conventions (exclusive vs. inclusive disjunction, quantifier phrasing) are followed [DATASET-D3, D11]. This strongly supports NL-only and hybrid NL+FOL evaluation modes. However, three concerns reduce the score: (1) The paper explicitly documents that input format materially affects performance — adding FOL helps GPT-4 but hurts GPT-3.5 [Q162, Q163] — meaning a pure-symbolic regime is poorly matched to FOLIO's documented score distributions; this is the live unresolved decision flagged by the lab. (2) Dataset analysis reveals concrete NL-FOL alignment failures: a direct negation drop creating semantic contradiction [DATASET-D17], an entity-name mismatch (John/Jim) [DATASET-D16], unbalanced parentheses in multiple FOL formulas [DATASET-D14, D15, D20, D21], and FOL over-specification (double existential where single suffices) [DATASET-D6]. (3) Subcorpus AST diversity differs (WikiLogic: 53; HybLogic: 33) [Q106], affecting structural variety in the symbolic layer.

ID	Evidence
Q37	"We also ask the annotators to write parallel FOL sentences for both the premises and conclusions." (p.4)
Q49	"Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly." (p.4)
Q50	"Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness." (p.4)
Q52	"We adopt the FOL definitions and syntax most widely used in the AI community (Russell and Norvig, 2010)." (p.5)
Q64	"Each natural language (NL) story S in FOLIO consists of n premises: P = {p1, p2, ..., pn} and m conclusions: H = {h1, h2, ..., hm}." (p.6)
Q65	"All NL stories are annotated with parallel FOL stories SF, which are sets of FOL formulas consisting of n premises PF = {pf1, pf2, ..., pfn} and m conclusions HF = {hf1, hf2, ..., hfm}." (p.6)
Q68	"In particular, each of the NL sentence pi or hi and the parallel FOL formula pfi or hfi should be logically and semantically equivalent." (p.6)
Q106	"In NL-FOL translation, performs 10 points better on HybLogic than WikiLogic. This could be because WikiLogic has more distinct and diverse sentence-level logical and language patterns and FOL annotations. WikiLogic has 53 ASTs while HybLogic has 33." (p.8)
Q162	"As shown in Table 11, the performance slightly increases in the NL+FOL setting for GPT-4 while GPT-3.5 performs worse in both the NL+FOL and the FOL-only settings." (p.15)
Q163	"FOL always serves as additional useful information for GPT-4, but not for GPT-3.5 regardless of whether FOL is concatenated with NL." (p.15)
D1	No plants are fungi. Mushrooms are fungi." / "∀x (Plant(x) → ¬Fungi(x)) ∀x (Mushroom(x) → Fungi(x)) — Simple 2-premise syllogism with exact NL/FOL parallel; confirms NL-first design with FOL annotation layer
D2	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises." / "∀x (InThisTechCompany(x) ∧ Consistent(x) ∧ StickTo(x, theirRegularRoutine) → ¬Like(x, surprise)) — Multi-premise story with 7 premises including mixed disjunction/implication; illustrates chain-depth complexity and NL-FOL alignment

D3	A project is written either in C++ or Python." / "∀x (Project(x) → (WrittenIn(x, cplusplus) ⊕ WrittenIn(x, python))) — Exclusive disjunction (⊕) expressed in NL as "either-or"; confirms documented NL convention alignment
D6	Some cats are not pets. All cats are mammals." / "∃x (Cat(x) ∧ ¬Pet(x)) ∀x (Cat(x) → Mammal(x)) — Minimal existential quantifier example; note that conclusion-FOL uses double existential (∃x ∃y) where the NL "some mammals" might only require ∃x — potential over-specification in FOL
D11	Everyone in the Franco-China diplomatic conference is either a PRC national or a French national, but not both." / "∀x (In(x, franco-ChinaDiplomaticConference) → PRCNational(x) ⊕ FrenchNational(x)) — NL "but not both" correctly mapped to ⊕; confirms NL disjunction conventions
D14	Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic error in training data
D15	∀x (MajorArgumentForm(x) → (InductiveReasoning(x) ⊕ DeductiveReasoning(x)) — Missing closing parenthesis in premises-FOL
D16	Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises." / "¬(ProfessionalBasketballPlayer(jim) ⊕ NeverExercises(jim)) — Premises describe "John" but FOL uses "jim" — name inconsistency between NL and FOL
D17	Matt does not invest in the public stock market regularly." / "InvestInRegularly(matt, publicStockMarket) — NL premise says Matt does NOT invest; FOL asserts he DOES — direct NL/FOL semantic contradiction
D20	(Knows(dan, dune) ∧ ScienceFiction(dune)) ∨ ProvedToBe(dune, false)) — Unbalanced parenthesis in premises-FOL
D21	Contain(tikTok, chatFeature) ⊕ ComputerProgram(tikTok)) — Unbalanced closing parenthesis in conclusion-FOL

Strengths

- Complete NL/FOL parallel annotation enables both NL-only and hybrid input modes out-of-the-box [DATASET-D1, D2]
- Documented NL conventions (⊕ for exclusive disjunction, ∨ for inclusive) consistently applied [DATASET-D3, D11]
- Standard AI/education FOL syntax (Russell & Norvig) used [Q52], aligning with lab researcher familiarity
- Grammar-reviewed NL with explicit ambiguity reduction protocols [Q49, Q50, Q51]

ID	Evidence
Q49	“Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly.” (p.4)
Q50	“Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness.” (p.4)
Q51	“We also eliminate natural language ambiguity when it is possible.” (p.4)
Q52	“We adopt the FOL definitions and syntax most widely used in the AI community (Russell and Norvig, 2010).” (p.5)
D1	No plants are fungi. Mushrooms are fungi." / "∀x (Plant(x) → ¬Fungi(x)) ∀x (Mushroom(x) → Fungi(x)) — Simple 2-premise syllogism with exact NL/FOL parallel; confirms NL-first design with FOL annotation layer
D2	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises." / "∀x (InThisTechCompany(x) ∧ Consistent(x) ∧ StickTo(x, theirRegularRoutine) → ¬Like(x, surprise)) — Multi-premise story with 7 premises including mixed disjunction/implication; illustrates chain-depth complexity and NL-FOL alignment
D3	A project is written either in C++ or Python." / "∀x (Project(x) → (WrittenIn(x, cplusplus) ⊕ WrittenIn(x, python))) — Exclusive disjunction (⊕) expressed in NL as "either-or"; confirms documented NL convention alignment
D11	Everyone in the Franco-China diplomatic conference is either a PRC national or a French national, but not both." / "∀x (In(x, franco-ChinaDiplomaticConference) → PRCNational(x) ⊕ FrenchNational(x)) — NL "but not both" correctly mapped to ⊕; confirms NL disjunction conventions

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs

IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal-distribution mismatch is the central concern. FOLIO's documented scores derive from NL-first presentation [Q64]; paper explicitly shows FOL-only and NL+FOL settings produce different results across models [Q162, Q163]. Pure-symbolic regime will not generalize to/from FOLIO's documented score distributions; hybrid NL+FOL is well-supported.

IF-2: Lab infrastructure (Python tooling, OpenAI API, standard NLP stack) fully supports all input formats; no infrastructure mismatch [elicitation infrastructure_notes].

IF-3: Domain-specific form differences relevant to the lab's pipeline: (a) NL/FOL semantic alignment failures observed in sample [DATASET-D17], (b) syntactic errors in FOL [DATASET-D14, D15, D20, D21], (c) entity-name inconsistencies [DATASET-D16], (d) FOL over-specification of existentials [DATASET-D6]. These corrupt any FOL-only or hybrid presentation that relies on FOL fidelity.

IF-4: Form mismatches that would harm external validity: (1) pure-symbolic regime is weakly aligned with FOLIO's NL-first design and validation; (2) NL/FOL annotation errors in the data create input-signal noise that may differentially affect hybrid vs. NL-only conditions; (3) AST diversity differs between subcorpora [Q106].

ID	Evidence
Q64	“Each natural language (NL) story S in FOLIO consists of n premises: P = {p1, p2, ..., pn} and m conclusions: H = {h1, h2, ..., hm}.” (p.6)
Q106	“In NL-FOL translation, performs 10 points better on HybLogic than WikiLogic. This could be because WikiLogic has more distinct and diverse sentence-level logical and language patterns and FOL annotations. WikiLogic has 53 ASTs while HybLogic has 33.” (p.8)
Q162	“As shown in Table 11, the performance slightly increases in the NL+FOL setting for GPT-4 while GPT-3.5 performs worse in both the NL+FOL and the FOL-only settings.” (p.15)
Q163	“FOL always serves as additional useful information for GPT-4, but not for GPT-3.5 regardless of whether FOL is concatenated with NL.” (p.15)
D6	Some cats are not pets. All cats are mammals." / "∃x (Cat(x) ∧ ¬Pet(x)) ∀x (Cat(x) → Mammal(x)) — Minimal existential quantifier example; note that conclusion-FOL uses double existential (∃x ∃y) where the NL "some mammals" might only require ∃x — potential over-specification in FOL
D14	Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic error in training data
D15	∀x (MajorArgumentForm(x) → (InductiveReasoning(x) ⊕ DeductiveReasoning(x)) — Missing closing parenthesis in premises-FOL
D16	Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises." / "¬(ProfessionalBasketballPlayer(jim) ⊕ NeverExercises(jim)) — Premises describe "John" but FOL uses "jim" — name inconsistency between NL and FOL
D17	Matt does not invest in the public stock market regularly." / "InvestInRegularly(matt, publicStockMarket) — NL premise says Matt does NOT invest; FOL asserts he DOES — direct NL/FOL semantic contradiction
D20	(Knows(dan, dune) ∧ ScienceFiction(dune)) ∨ ProvedToBe(dune, false)) — Unbalanced parenthesis in premises-FOL
D21	Contain(tikTok, chatFeature) ⊕ ComputerProgram(tikTok)) — Unbalanced closing parenthesis in conclusion-FOL
WEB-5	Auto-formalization agent 86.70% on cleaned FOLIO vs. 73.89% CoT baseline (https://arxiv.org/pdf/2603.02788)

Information Gaps

- Rate of NL/FOL annotation errors across full corpus not quantified
- Whether the Stanford CS221 inference engine successfully resolved FOL syntactic errors via the Python parser bridge, or whether they propagated to label errors
- How pure-symbolic FOL-only presentation would interact with the observed FOL annotation errors

Requires Expert Verification

- Whether the lab should use the original FOLIO dataset or the 2025 cleaned validation set
- Whether observed FOL annotation errors are present in the test split (not accessible via HuggingFace)

Remediation

Gap 1: FOLIO is NL-first; pure-symbolic regime is weakly aligned with documented score distributions, and NL/FOL annotation errors observed in sample data corrupt FOL-only and hybrid signals

Recommendation: Restrict primary FOLIO evaluation to NL-only or hybrid NL+FOL conditions. For pure-symbolic evaluation, use ProverQA (ICLR 2025) or LogicGraph as primary; treat FOLIO FOL-only as ablation only. Apply the 2025 Vampire-cleaned annotation set (or equivalent QA pass) before any FOL-input experiments.

Output Ontology (OO)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US AI Research Lab — FOLIO FOL Reasoning Evaluation*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology is an exact match for the lab's deployment. FOLIO's three-way labels (True, False, Unknown) [Q36, Q66] map cleanly to the lab's three categories (entailment, contradiction, neutral) per elicitation Q1, with no sub-distinctions needed. The taxonomy is grounded in objective FOL inference rather than annotator subjectivity [Q2, Q70]. The label distribution in the test set is near-balanced (87 True / 78 False / 61 Unknown, 38.5% majority baseline) [Q94]. Dataset analysis confirms the Unknown label is used with logical precision rather than as a catch-all — complementary conclusions ('Harry is cool' and 'Harry is not cool') are both correctly labeled Uncertain when premises underdetermine them [DATASET-D19]. The only minor concern is the HuggingFace dataset uses the string 'Uncertain' rather than 'Unknown' [DATASET-D2], requiring a trivial string normalization. Temporal/modal exclusions [Q139, Q140] match the lab's out-of-scope list exactly.

ID	Evidence
Q1	“FOLIO consists of 1,430 examples (unique conclusions), each paired with one of 487 sets of premises used to deductively reason for the validity of each conclusion.” (p.1)
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q36	“Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example).” (p.4)
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q70	“In our dataset, the premises and conclusions are set up in such a way to ensure that the inference engine always returns an answer given enough resources such as time and memory.” (p.6)
Q94	“The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively.” (p.7)
Q139	“Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics.” (p.13)
Q140	“Consequently, they are beyond the scope of the definition of first-order logic used in our dataset.” (p.13)
D2	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises." / "∀x (InThisTechCompany(x) ∧ Consistent(x) ∧ StickTo(x, theirRegularRoutine) → ¬Like(x, surprise)) — Multi-premise story with 7 premises including mixed disjunction/implication; illustrates chain-depth complexity and NL-FOL alignment
D19	Same story_id=482 (Potterville), three conclusions: "Harry is cool" (Uncertain), "Harry is not cool" (Uncertain), "Harry is a wizard or angry" (False) — Both "Harry is cool" and "Harry is not cool" labeled Uncertain — confirms Unknown/Uncertain label calibration; also Harry Potter cultural reference

Strengths

- Three-way label maps directly to lab's entailment/contradiction/neutral with no residual ambiguity (confirmed by elicitation Q1)
- Ground truth determined by automated FOL inference engine, not annotator agreement [Q2, Q70]
- Near-balanced label distribution in test set [Q94]
- Empirical verification that Unknown reflects logical underdetermination [DATASET-D19]

ID	Evidence
Q1	“FOLIO consists of 1,430 examples (unique conclusions), each paired with one of 487 sets of premises used to deductively reason for the validity of each conclusion.” (p.1)
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q70	“In our dataset, the premises and conclusions are set up in such a way to ensure that the inference engine always returns an answer given enough resources such as time and memory.” (p.6)
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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output categories (True/False/Unknown) are precisely relevant to the lab's three-way classification regime; elicitation Q1 confirmed clean mapping.

OO-2: No missing regional categories. The lab confirmed no finer-grained distinctions within Unknown are needed.

OO-3: No categories encode non-regional or value-laden assumptions; labels are anchored in formal logical validity [Q2, Q137].

OO-4: No stakeholder-driven redesign needed; taxonomy is formally objective.

OO-5: No taxonomy issues. Minor implementation note: HF label string is 'Uncertain' not 'Unknown' [DATASET-D2] — string normalization only. 5% of conclusions have complex syntactic structure posing comprehension challenges [Q120] but this is a difficulty property, not a taxonomy issue.

ID	Evidence
Q1	“FOLIO consists of 1,430 examples (unique conclusions), each paired with one of 487 sets of premises used to deductively reason for the validity of each conclusion.” (p.1)
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q36	“Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example).” (p.4)
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q70	“In our dataset, the premises and conclusions are set up in such a way to ensure that the inference engine always returns an answer given enough resources such as time and memory.” (p.6)
Q120	“5% of conclusions in FOLIO have a complex syntactic structure, posing comprehension challenges for GPT-4.” (p.9)
Q137	“FOL has no ambiguity while ambiguity can occur at various levels of NLP.” (p.13)

D2	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises." / "∀x (InThisTechCompany(x) ∧ Consistent(x) ∧ StickTo(x, theirRegularRoutine) → ¬Like(x, surprise)) — Multi-premise story with 7 premises including mixed disjunction/implication; illustrates chain-depth complexity and NL-FOL alignment

Output Content (OC)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US AI Research Lab — FOLIO FOL Reasoning Evaluation*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (label correctness) is anchored in objective formal logical validity rather than annotator subjectivity, eliminating most convergent-validity concerns for the primary task [Q2, Q57, Q153]. A rigorous multi-stage QA process was used: NL quality review separated from FOL quality review [Q29, Q124, Q125], standardized FOL annotation protocol [Q54, Q143-Q146], commonsense premises explicitly added at alignment stage [Q55], and 39.2% of stories rewritten in QA [Q48]. Expert/non-expert accuracy gap (95.98% vs. 61.82%) [Q111] confirms the task genuinely requires FOL expertise and the labels are not noise. However, dataset analysis surfaces concrete label-correctness concerns: (1) a direct NL/FOL semantic contradiction in Example 57 (negation dropped) [DATASET-D17] means the label may be computed from incorrect FOL; (2) syntactic errors in FOL formulas [DATASET-D14, D15, D20, D21] could corrupt inference engine output; (3) FOL over-specification (double existential) [DATASET-D6] and atypical formulations [DATASET-D25] suggest annotation-protocol inconsistencies. A 2025 study using the Vampire prover identified errors requiring a cleaned validation set [WEB-5]. Annotator demographics beyond English proficiency and FOL training are not documented [Q122, Q123] — relevant if this were a subjective task, but largely moot here since labels are inference-engine-derived.

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q29	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.4)
Q48	"Toward these objectives, we manually screened all stories and found that 39.2% of the stories suffer from at least one of these issues. We implemented a detailed protocol to rewrite these stories." (p.4)
Q54	"One NL sentence can be translated into FOL through multiple non-equivalent ways. For example, sometimes additional information inferred from a sentence can be represented in FOL, leading to multiple representations. We therefore design an annotation protocol for FOL translation in order to ensure that our FOL translations are as consistent as possible across all examples in our dataset." (p.5)
Q55	"Apart from checking whether NL and FOL express equivalent meanings, we also add necessary commonsense knowledge in both the NL and FOL premises. Sometimes humans do not write certain commonsense knowledge in the premises that is required in the FOL reasoning process, which is based solely on the premises given. We add such knowledge as additional premises at this stage." (p.5)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q122	"Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.12)
Q123	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing." (p.12)
Q124	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved." (p.12)
Q125	"For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.12)
Q143	"We therefore design an annotation protocol for first-order logic translation in order to ensure that our FOL translations are as consistent as possible across all examples in our dataset." (p.14)

Q144	"First-order logic formulas need to preserve as much as possible the semantics of natural language sentences." (p.14)
Q145	"First-order logic formulas should stay as faithful to the structure of the original NL sentence as possible." (p.14)
Q146	"Semantic decomposition is not needed unless necessary for maintaining the NL expressivity." (p.14)
Q153	"Proving a story requires three steps. First, the FOL statements of the premises and conclusions of a story annotated by humans are converted to Python code. Then, the code snippets are used as input to the theorem prover. Finally, the theorem prover outputs whether the conclusions are True / False / Unknown, based on the premises." (p.14)
D6	Some cats are not pets. All cats are mammals." / "∃x (Cat(x) ∧ ¬Pet(x)) ∀x (Cat(x) → Mammal(x)) — Minimal existential quantifier example; note that conclusion-FOL uses double existential (∃x ∃y) where the NL "some mammals" might only require ∃x — potential over-specification in FOL
D14	Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic error in training data
D15	∀x (MajorArgumentForm(x) → (InductiveReasoning(x) ⊕ DeductiveReasoning(x)) — Missing closing parenthesis in premises-FOL
D17	Matt does not invest in the public stock market regularly." / "InvestInRegularly(matt, publicStockMarket) — NL premise says Matt does NOT invest; FOL asserts he DOES — direct NL/FOL semantic contradiction
D20	(Knows(dan, dune) ∧ ScienceFiction(dune)) ∨ ProvedToBe(dune, false)) — Unbalanced parenthesis in premises-FOL
D21	Contain(tikTok, chatFeature) ⊕ ComputerProgram(tikTok)) — Unbalanced closing parenthesis in conclusion-FOL
D25	Conclusion-FOL: ∃x (SavesFor(jane, enoughMoney, theSummer) ∧ Kangaroo(x) → WillSee(x, jane, berlinzoo)) — Conclusion-FOL encodes conditional inside existential scope — unusual FOL formulation that may represent annotation inconsistency
WEB-5	Auto-formalization agent 86.70% on cleaned FOLIO vs. 73.89% CoT baseline (https://arxiv.org/pdf/2603.02788)

Strengths

- Labels objectively determined by automated FOL inference engine, not subjective judgment [Q2, Q153]
- Rigorous multi-stage QA pipeline with role-separated expert review [Q29, Q124, Q125]
- Standardized FOL annotation protocol to minimize translation inconsistencies [Q54, Q143-Q146]
- Expert human ceiling (95.98%) provides meaningful upper bound [Q111]

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q29	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.4)
Q54	"One NL sentence can be translated into FOL through multiple non-equivalent ways. For example, sometimes additional information inferred from a sentence can be represented in FOL, leading to multiple representations. We therefore design an annotation protocol for FOL translation in order to ensure that our FOL translations are as consistent as possible across all examples in our dataset." (p.5)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q124	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved." (p.12)
Q125	"For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.12)
Q143	"We therefore design an annotation protocol for first-order logic translation in order to ensure that our FOL translations are as consistent as possible across all examples in our dataset." (p.14)
Q144	"First-order logic formulas need to preserve as much as possible the semantics of natural language sentences." (p.14)
Q145	"First-order logic formulas should stay as faithful to the structure of the original NL sentence as possible." (p.14)
Q146	"Semantic decomposition is not needed unless necessary for maintaining the NL expressivity." (p.14)
Q153	"Proving a story requires three steps. First, the FOL statements of the premises and conclusions of a story annotated by humans are converted to Python code. Then, the code snippets are used as input to the theorem prover. Finally, the theorem prover outputs whether the conclusions are True / False / Unknown, based on the premises." (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth reflects formal logical validity rather than stakeholder perspective; the lab's expert population is precisely the population whose judgments the FOL-engine-derived labels approximate [Q2, Q111].

OC-2: Minimal disagreement risk: ground truth is mechanically derived from FOL premises, and the lab's expert population shares the source culture (FOL-trained CS researchers). Expert annotators achieved 95.98% on FOLIO [Q111].

OC-3: Annotator demographics partially documented: native/near-native English, CS students/researchers with formal FOL education [Q27, Q28, Q122, Q123]. Gender, racial, and institutional diversity NOT documented — INSUFFICIENT DOCUMENTATION, but largely moot given objective grounding.

OC-4: Not needed — labels are inference-engine-derived; regional annotator pool re-annotation would not change formal-logical ground truth.

OC-5: Aggregation methods not relevant — single ground-truth label per conclusion derived from FOL prover, not majority vote.

OC-6: Label issues identified: (1) NL/FOL semantic mismatch in Example 57 [DATASET-D17] — if label was computed from FOL, it may not match what NL premises imply; (2) syntactic errors in FOL [DATASET-D14, D15, D20, D21] may have corrupted inference engine output or required manual override; (3) 2025 study found errors warranting cleaning [WEB-5].

ID	Evidence
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q28	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-” (p.3)
Q70	“In our dataset, the premises and conclusions are set up in such a way to ensure that the inference engine always returns an answer given enough resources such as time and memory.” (p.6)
Q111	“Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%.” (p.9)
Q122	“Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.12)
Q123	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing.” (p.12)
Q153	“Proving a story requires three steps. First, the FOL statements of the premises and conclusions of a story annotated by humans are converted to Python code. Then, the code snippets are used as input to the theorem prover. Finally, the theorem prover outputs whether the conclusions are True / False / Unknown, based on the premises.” (p.14)
D14	Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic error in training data
D15	∀x (MajorArgumentForm(x) → (InductiveReasoning(x) ⊕ DeductiveReasoning(x)) — Missing closing parenthesis in premises-FOL

D17	Matt does not invest in the public stock market regularly." / "InvestInRegularly(matt, publicStockMarket) — NL premise says Matt does NOT invest; FOL asserts he DOES — direct NL/FOL semantic contradiction
D20	(Knows(dan, dune) ∧ ScienceFiction(dune)) ∨ ProvedToBe(dune, false)) — Unbalanced parenthesis in premises-FOL
D21	Contain(tikTok, chatFeature) ⊕ ComputerProgram(tikTok)) — Unbalanced closing parenthesis in conclusion-FOL
WEB-5	Auto-formalization agent 86.70% on cleaned FOLIO vs. 73.89% CoT baseline (https://arxiv.org/pdf/2603.02788)

Information Gaps

- Rate of label-corruption due to NL/FOL annotation errors across the full corpus
- Whether the inference engine produced labels using corrected FOL (post-QA) or whether visible FOL errors in the HF data correspond to label-affecting errors
- Annotator demographics beyond English proficiency and FOL training

Requires Expert Verification

- Whether lab uses original or 2025-cleaned FOLIO annotations
- Sample-based audit of label correctness against an independent FOL prover (e.g., Vampire or Prover9) to estimate label-noise rate

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Remediation

Gap 1: Direct NL/FOL semantic contradictions and syntactic errors observed in sampled training data; 2025 study confirmed corpus-wide annotation errors warranting cleaning

Recommendation: Use the 2025 cleaned FOLIO validation set (or run an independent Vampire/Prover9 pass to identify and exclude examples with corrupted FOL or NL/FOL mismatches). Report results on both cleaned and original annotation sets to allow comparison with published baselines.

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US AI Research Lab — FOLIO FOL Reasoning Evaluation*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form aligns well for the primary task: classification accuracy on three-way categorical labels [Q80] matches the lab's exact intended metric. Premise ordering has been verified not to be exploitable as a positional heuristic [Q158, Q159, Q160], reinforcing metric robustness. For the secondary NL-FOL translation task, both Syntactic Validity (SynV) [Q73, Q74] and Inference Engine Execution Accuracy (ExcAcc) [Q75, Q76] are defined separately, supporting clean task separation. However, three concerns prevent a 5: (1) The original GPT-4 ceiling (53.1% on HybLogic) [Q20] is now substantially closed by neuro-symbolic pipelines (LLM-ARC 88.32% vs. human 95.98%) [WEB-4], creating a discriminability bifurcation between standard prompting and neuro-symbolic regimes — ceiling concern is reduced but remains live for standard prompting [WEB-4]. (2) Systematic label-class asymmetry: accuracy on False/Unknown is 54.0% few-shot vs. 61.9% fine-tuning, much lower than True [Q155, Q156, Q157] — the lab must account for this when interpreting aggregate accuracy. (3) Test split is not present in the HuggingFace dataset [DATASET concern 4]; only train (1,001) + validation (203) are available, reducing statistical power for primary evaluations on the documented test split (226 examples).

ID	Evidence
Q20	“Under the few-shot setting, the most capable publicly available LLM so far achieves only 53.1% on the stories written in a hybrid manner, which is slightly better than random.” (p.2)
Q73	“Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV).” (p.6)
Q74	“The Syntactic Validity score measures whether the FOL formulas are syntactically valid. The score will be 1 if all FOL formulas of an example can pass the syntactic check and 0 otherwise 2).” (p.6)
Q75	“Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion.” (p.6)
Q76	“We define the accuracy of the output labels as the execution accuracy.” (p.6)
Q80	“We use accuracy for evaluating logical reasoning performance.” (p.6)
Q155	“Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown.” (p.15)
Q156	“The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting.” (p.15)
Q157	“They also tend to make more predictions of True than the other labels.” (p.15)
Q158	“To test if the premise ordering in FOLIO has spurious correlations with the conclusion label which a model can exploit, we shuffle the input premises to evaluate models.” (p.15)
Q159	“We find that accuracy increases or decreases by roughly 1% in most settings compared to our unshuffled premises.” (p.15)
Q160	“This indicates that the ordering of premises in FOLIO examples does not yield significant information about the label, and thus models will not be able to use the premise ordering as a strong heuristic or statistical feature for its predictions.” (p.15)
WEB-4	LLM-ARC neuro-symbolic system achieves 88.32% on FOLIO validation (2024 SotA) (https://arxiv.org/pdf/2406.17663)

Strengths

- Classification accuracy metric is an exact match for lab's intended evaluation regime [Q80]
- Premise-ordering robustness empirically verified [Q158-Q160]

- Separate SynV and ExcAcc metrics enable clean task separation for NL-FOL translation [Q73-Q76]
- Human expert ceiling (95.98%) provides interpretable upper bound [Q111, Q113]

ID	Evidence
Q73	“Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV).” (p.6)
Q74	“The Syntactic Validity score measures whether the FOL formulas are syntactically valid. The score will be 1 if all FOL formulas of an example can pass the syntactic check and 0 otherwise 2).” (p.6)
Q75	“Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion.” (p.6)
Q76	“We define the accuracy of the output labels as the execution accuracy.” (p.6)
Q80	“We use accuracy for evaluating logical reasoning performance.” (p.6)
Q111	“Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%.” (p.9)
Q113	“The expert and GPT-4 gap is 31.82%, suggesting significant room for model improvement.” (p.9)
Q158	“To test if the premise ordering in FOLIO has spurious correlations with the conclusion label which a model can exploit, we shuffle the input premises to evaluate models.” (p.15)
Q159	“We find that accuracy increases or decreases by roughly 1% in most settings compared to our unshuffled premises.” (p.15)
Q160	“This indicates that the ordering of premises in FOLIO examples does not yield significant information about the label, and thus models will not be able to use the premise ordering as a strong heuristic or statistical feature for its predictions.” (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality matches exactly: categorical three-way label for primary task [Q80]; SynV/ExcAcc for secondary translation task [Q73-Q76]. Both align with lab's stated metrics.

OF-2: Not applicable — no speech-based output requirements in this deployment.

OF-3: Not applicable — expert evaluator population in well-resourced lab environment.

OF-4: Form mismatches: (a) HF dataset lacks test split [DATASET concern 4]; primary documented performance comparisons reference the unavailable 226-example test set, so lab must obtain it from FOLIO GitHub or use validation set with reduced power (n=203). (b) Ceiling concern is partially resolved for neuro-symbolic pipelines [WEB-4] but remains live for standard prompting baselines on frontier models (GPT-4o, Claude 3.5, Gemini 1.5) which are not publicly documented [WEB-4]. (c) Systematic label-class asymmetry [Q155-Q157] means aggregate accuracy can mislead — per-class reporting is needed.

ID	Evidence
Q20	“Under the few-shot setting, the most capable publicly available LLM so far achieves only 53.1% on the stories written in a hybrid manner, which is slightly better than random.” (p.2)
Q73	“Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV).” (p.6)
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WEB-5	Auto-formalization agent 86.70% on cleaned FOLIO vs. 73.89% CoT baseline (https://arxiv.org/pdf/2603.02788)

Information Gaps

- Vanilla prompting baselines for GPT-4o, Claude 3.5, Gemini 1.5 on FOLIO not publicly documented [WEB-4]
- Per-depth example counts at depth 6–7 — likely too small for reliable per-depth accuracy estimates [WEB-6]
- Test set availability via HuggingFace — must be sourced from GitHub

Requires Expert Verification

- Whether the lab's specific evaluation paradigm (standard prompting vs. neuro-symbolic) maintains discriminability for its target frontier-model set
- Statistical power analysis for per-depth and per-label-class metrics on the lab's chosen evaluation split

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Remediation

Gap 1: HuggingFace dataset lacks the documented 226-example test split; using only validation (n=203) reduces statistical power for per-depth and per-label-class analyses

Recommendation: Source the 226-example test split directly from the FOLIO GitHub repository. Report per-label-class accuracy (True / False / Unknown) separately given the documented systematic asymmetry [Q155-Q157], and treat per-depth scores at depth 6–7 as indicative only due to small subset sizes.

Gap 2: Standard-prompting ceiling for frontier models (GPT-4o, Claude 3.5, Gemini 1.5) on FOLIO is not publicly documented; neuro-symbolic SotA is 88.32% vs. human 95.98%, leaving discriminability uncertain for frontier prompting

Recommendation: Before deploying FOLIO as a discriminative benchmark, run a calibration pass on the lab's target frontier models with the lab's chosen prompting strategy. If ceiling effects emerge, supplement with LogicGraph or LogiConBench for harder depth tiers.

ID	Evidence
Q155	“Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown.” (p.15)
Q156	“The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting.” (p.15)
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